

AIBS - Research Proposal (first draft)

Handbook, Group 3: Tanishq Bhajji, Nikola Hadzhiminov, Valentina Petrušić, Tanjin Rahman Rafi, Anisah Sakura Sutrisno

1. **Target audience:** Our target audience is the elderly care firm STMG in Arnhem. It operates in the healthcare sector, with a size of approximately 4.500 clients and around 1.200 staff. The employees have to make work schedules by making phone calls and the elderly are not informed well enough who and when is going to take care of them and have to manually call workers. This creates an uncertainty and extra workload on the people managing the phone calls. The question is how can AI facilitate distribution planning and help the staff and the elderly.
2. **What they need:**
The SME needs to get comfortable with using AI-tools, know what the costs are, what the AI-tool can help with and to train employees to gain an understanding of this new AI-tool/system within the workspace. It is important to keep the human always in the loop. Even a simple AI-supported facilitation of an SME can significantly improve worker efficiency. Thus, improving productivity.
3. **Research question.**
Main question: Should STMG adopt an AI-supported tool to give clients advance notice of caretaker visits, and what would adopting one actually require?

External analysis

1. What AI-driven scheduling/notification tools already exist for home-care or field-service logistics, and how mature and affordable are they for a Dutch not-for-profit care organisation?
2. What do GDPR, health-adjacent data rules, and WMO/WLZ funding conditions require or restrict for a tool that touches elderly clients' schedules and location data?

Internal analysis

1. How does STMG currently plan and communicate caretaker visits to clients and staff, and at what point does the information break down?
2. What digital systems, skills and demand for AI exist today among STMG's schedulers and caretakers?
3. How would STMG's elderly clients (and their families) actually want to be notified, e.g via app, SMS, phone call, and what does that imply for adoption

risk?

4. **Quality criteria for the handbook.**

- The handbook mirrors our AEL platform's output qualities
- It is clear to a non-technical owner
- It is accurate against sources we've appraised
- Honesty about responsible AI and trust issues (especially with elderly, health-adjacent client data) is upheld.
- It is useful enough that STMG's management could act on it
- A chapter in the handbook is included that ends in concrete first steps rather than general commentary.

5. **Methodology.**

We will use the Desk research that serves questions 1 and 2 mainly: CBS data on home-care staffing and digitisation, STMG's own Chamber of Commerce record, ActiZ (the sector body) and other branch publications, regional Gelderland press on care-sector digitisation, and vendor material for existing scheduling/notification tools used in home care or similar field-service settings.

Field research will include one visit to STMG in weeks 2 - 4, serving questions 3, 4 and 5. We will plan an interview with a planner and, if possible, caretakers. A team member who works with STMG can help us contact the right people and some elderly clients. We will also interview a few elderly clients to understand how they prefer to receive visit updates, for example by SMS, phone call or app. This will help us understand the current problems and whether an AI-supported tool would be useful for STMG.